Fault diagnosis of rotating machinery is a fundamental task in industrial maintenance. Deep learning has been widely adopted for this purpose over the past decade, and many variants have been proposed for vibration classification.

TABLE I
FAULTS OF THE TEST DATASET

Type	No. of samples	Information of Faults
1	391	K Scatch
2	402	Bumps
3	673	Other Faults
4	1204	Inner Race

The dataset contains four classes of faults collected from a test rig. Each sample is a vibration signal of 2048 points sampled at 12 kHz. The signals are all normalized before model training begins.

A. Model Training

To make the agent network converge better, each probability of the pseudo-labels is initialized to 0.5. The inverse class frequency is used when the class distribution is quite different.

ALGORITHM 1

Input: labeled set Dl and unlabeled set Du
1 Pre-train TPG following Eq.1.
2 Initialize s0 by new sequence.
3 for t = 0 to T do
4 Sample the pseudo-label
5 end

The reward function in the environment can then be calculated as follows.

$$rt = R(Yv, Yt) \tag{11}$$

where rt denotes the reward at iteration t and R denotes the evaluation metric. Then the new state can be obtained from the environment directly.

C. Optimization of the MDP

After all components are defined, an appropriate optimization algorithm is required to train the agent. The reward function is nondifferentiable.

$$pi(at|st) = \frac{exp(dt)}{sum\ exp(dj)} \tag{12}$$

where dt is the score of action t. The softmax denominator sums over all admissible actions.

TABLE II
VECTOR TABLE WITHOUT TEXT LAYER

methods. Thus, Self-Training-SVM and TPG-SVM can obtain relatively good results. Consequently, all of the above analysis indicates that TPG-SVM has a high ability in dealing with fault classification.